

Yahya Mirza

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EDUCATION

University of California, San Diego

Bachelor of Science in Mechanical Engineering

La Jolla, CA

Sep. 2025 – June 2027

Relevant Coursework: Fluid Mechanics, Thermodynamics, Solid Mechanics, Statics, Dynamics, Linear Circuits, Machining, MATLAB, Heat Transfer, Numerical Methods, Partial Differential Equations, SOLIDWORKS, Linear Algebra, Vector Calculus

Foothill College

Mechanical Engineering for Transfer (GPA: 4.0)

Los Altos Hills, CA

Sep. 2024 – Aug. 2025

SKILLS

Mechanical Design · SOLIDWORKS · GD&T · OnShape · Fusion 360 · Arduino · ESP32 · Raspberry Pi · C++ · MATLAB · FEA Analysis · Machining · GitHub · CNC Milling · Risk Reduction · Technical Writing · AutoCAD · DFM/DFA · Soldering · Kenesto

EXPERIENCE

Mechanical Lead – Yonder Dynamics (UCSD's Competition Mars Rover Team)

Sep. 2025 – Present

University of California, San Diego

La Jolla, CA

- Led end-to-end production of a **linear slider mechanism** capable of reliably translating a 30lb, 5-axis robotic arm with a **range of motion of 10"**.
- Developed a parametric **CAD model** in **OnShape**, implementing **DFM/DFA** procedures to ensure cheap and quick fabrication.
- Verified via **static analysis calculations** (with **5x FOS**) that the module powered with a NEO550 brushless motor coupled with a 50:1 Harmonic Drive gearbox can reliably translate a **200N payload** at 60 degrees from horizontal.
- Fabricated over **20+ parts** (sprockets, housings, structural members) via **milling (manual and CNC), water jetting, 3D printing, laser cutting, and sand blasting**.
- Planned **validation activities** to determine **max load, suitable operation temperatures, and run-time**.
- Documented the project in an **engineering report** containing detailed diagrams, **FBDs**, calculations, and design iterations.

Driver Interface and Safety Technician – Triton Baja (UCSD's Baja SAE Team)

June 2026 – Present

University of California, San Diego

La Jolla, CA

- Developed a **MATLAB script** to **predict optimal positions** for the driver seat and brake pedal given prospective driver calf and thigh lengths.
- Compiled driver data to determine **final brake pedal placement**. Created a rough CAD model of the **driver pedal box** in SOLIDWORKS, now optimizing via iterative FEA.
- Analyzed pedal-box constraints via a **weighted decision matrix** (brake ratio, mounting, packaging, cost, machining) to settle on a **top-mounted pedal box design**.

PROJECTS

Whiteboard Drawing Robot | Assembly Level Design, OnShape, Arduino, Electromechanical Systems

Jul 2026 – Present

- Conducted design, research, and production of a **two-axis, belt-driven gantry system** to accurately position a marker within a **40cm x 50cm whiteboard**.
- Maintained a **top-down, assembly driven approach** to CAD in **OnShape** (Geometry Planning Assembly → Develop Key Mechanisms → Develop Structural Components → Final Assembly) totaling over **30 unique 3D printed parts** within a **singular Part Studio**.
- Structured CAD to be parametric, creating variables when necessary. Organized features by each distinct part, limiting feature dependencies to primarily exist within dedicated folders.
- Developed **test circuitry** to **tune/configure belt tension, speed, and runtime** via **Arduino**, universal microstep drivers, a **DC power supply**, and **potentiometers**. Need to develop Z-axis before attempting advanced software control.

Competition Soccer Robot | Fusion 360, Statics, Machining, Circuitry

Oct 2025 – Dec 2025

- Successfully built a competition robot designed to sweep and score 4" dia. soccer balls into opponent's goal. Developed a **two-stage, belt-driven flywheel kicker mechanism (9:1 reduction)** resulting in an average **kicking distance** of **70%** of the span of the field, **rapidly iterating** via risk reduction procedures.
- Developed a comprehensive CAD model in **Fusion 360**, notably **designing** the robot's **swiping arm subassembly**.
- **Validated** the arm module with a **6x FOS** via static analysis and **experimental verification**, authoring both a **technical paper** and **presentation** documenting results.
- **Awarded** for **"Best Manufacturing"**; chosen by engineering professors and competition judges.